

ELEC 221 Tutorial 4 – Solutions

The Fourier Series and Its Properties

Monday, 5 October 2026

1. The Fourier series
2. Properties of Fourier series

1.1 – Deriving a_0 , a_k , b_k : setup

Start from $x(t) = a_0 + \sum_{k=1}^{\infty} (a_k \cos(k\omega t) + b_k \sin(k\omega t))$. The derivation relies on **orthogonality** over one period T ($\omega = 2\pi/T$):

$$\int_T \cos(k\omega t) dt = \int_T \sin(k\omega t) dt = 0 \quad (k \geq 1)$$

$$\int_T \cos(k\omega t) \cos(m\omega t) dt = \int_T \sin(k\omega t) \sin(m\omega t) dt = \frac{T}{2} \delta_{km}$$

$$\int_T \sin(k\omega t) \cos(m\omega t) dt = 0 \quad (\text{always})$$

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For a_0 : integrate both sides over T . Every cos/sin term vanishes, leaving $\int_T x(t) dt = a_0 T$, so

$$a_0 = \frac{1}{T} \int_T x(t) dt$$

1.1 – Deriving a_k and b_k

For a_m ($m \geq 1$): multiply both sides by $\cos(m\omega t)$ and integrate over T . We use m here (instead of k) purely to avoid clashing with the summation index k already running inside $\sum_k a_k \cos(k\omega t)$. By orthogonality every term vanishes except the $\cos(k\omega t) \cos(m\omega t)$ term with $k = m$:

$$\int_T x(t) \cos(m\omega t) dt = a_m \cdot \frac{T}{2} \implies a_m = \frac{2}{T} \int_T x(t) \cos(m\omega t) dt$$

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Since m was an arbitrary positive integer, relabel $m \rightarrow k$ to match the notation used everywhere else (Eq. (4), the problem statement):

$$a_k = \frac{2}{T} \int_T x(t) \cos(k\omega t) dt, \quad b_k = \frac{2}{T} \int_T x(t) \sin(k\omega t) dt$$

1.2 – Even/odd signals $\Rightarrow$ even/odd (real/imaginary) coefficients

For any real $x(t)$, conjugate symmetry always holds: $c_{-k} = c_k^*$. We now also use the substitution $t \rightarrow -t$ inside the analysis equation (over a symmetric interval $-T/2$ to $T/2$):

$$c_{-k} = \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{jk\omega t} dt \stackrel{t \rightarrow -t}{=} \frac{1}{T} \int_{-T/2}^{T/2} x(-t) e^{-jk\omega t} dt$$

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If $x(t)$ is even ($x(-t) = x(t)$): the integral becomes exactly c_k , so $c_{-k} = c_k$. Combined with $c_{-k} = c_k^*$: $c_k = c_k^* \Rightarrow$ c_k is real, and $c_{-k} = c_k$ (even in k).

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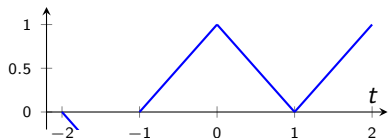
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If $x(t)$ is odd ($x(-t) = -x(t)$): the integral becomes $-c_k$, so $c_{-k} = -c_k$ (odd in k). Combined with $c_{-k} = c_k^*$: $-c_k = c_k^*$, which means the real part of c_k is zero, i.e. c_k is purely imaginary, and $c_{-k} = -c_k$.

1.3 – Triangular wave: qualitative analysis

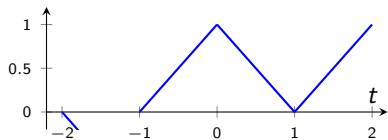
$$x(t) = \begin{cases} t + 1, & -1 < t \leq 0 \\ 1 - t, & 0 < t \leq 1 \end{cases}, \quad T = 2, \quad \omega = \pi$$



- $x(t)$ is **real and continuous** (no jumps – the two linear pieces meet at 0 and ± 1).
- $x(t)$ is **even**: $x(-t) = x(t)$ (check e.g. $x(0.5) = x(-0.5) = 0.5$).

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Conclusion: by 1.2, a real even signal has $b_k = 0$ for all k . So Eq. (4) (the sin/cos form) is more convenient here – we only need to compute a_0 and a_k , and can skip the b_k integrals entirely.

1.3 – Computing a_0 and a_k

$$a_0 = \frac{1}{T} \int_{-1}^1 x(t) dt = \frac{1}{2} \left[\int_{-1}^0 (t+1) dt + \int_0^1 (1-t) dt \right] = \frac{1}{2}(0.5 + 0.5) = \frac{1}{2}$$

(matches the average height of a triangle of base 2, peak 1: $\text{area}/T = 1/2$).

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$$a_k = \frac{2}{T} \int_{-1}^1 x(t) \cos(k\pi t) dt = \int_{-1}^1 x(t) \cos(k\pi t) dt = 2 \int_0^1 (1-t) \cos(k\pi t) dt$$

(even integrand, so double the integral over $[0, 1]$). Splitting and integrating by parts:

$$\int_0^1 \cos(k\pi t) dt = 0, \quad \int_0^1 t \cos(k\pi t) dt = -\frac{1 - \cos(k\pi)}{(k\pi)^2}$$

$$a_k = 2 \cdot \frac{1 - \cos(k\pi)}{(k\pi)^2} = \frac{2(1 - (-1)^k)}{(k\pi)^2}$$

1.3 – Result

Since $1 - (-1)^k$ is 0 for even k and 2 for odd k :

$$a_0 = \frac{1}{2}, \quad a_k = \begin{cases} \frac{4}{k^2\pi^2}, & k \text{ odd} \\ 0, & k \text{ even} \end{cases}, \quad b_k = 0 \text{ (all } k)$$

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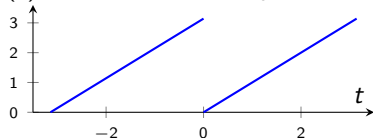
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Equivalently, in complex-exponential form (since x is real and even, c_k is real and $c_k = c_{-k} = a_k/2$ for $k \neq 0$):

$$c_0 = \frac{1}{2}, \quad c_k = \begin{cases} \frac{2}{k^2\pi^2}, & k \text{ odd} \\ 0, & k \text{ even} \end{cases}$$

1.4 – Sawtooth wave

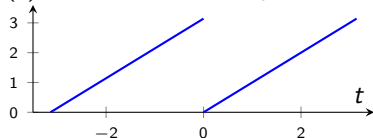
$x(t) = t$ for $0 \leq t < \pi$, period $T = \pi$, $\omega = 2\pi/T = 2$.



Neither even nor odd (jump at the period boundary), so we compute c_k directly rather than hunting for a symmetry shortcut.

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$$c_0 = \frac{1}{\pi} \int_0^{\pi} t \, dt = \frac{1}{\pi} \cdot \frac{\pi^2}{2} = \frac{\pi}{2}$$

For $k \neq 0$: $c_k = \frac{1}{\pi} \int_0^{\pi} t e^{-j2kt} \, dt$. Integrate by parts with $u = t$, $dv = e^{-j2kt} \, dt$, so $v = \frac{j}{2k} e^{-j2kt}$.

1.4 – Integration by parts

$$\int_0^{\pi} t e^{-j2kt} dt = \underbrace{\left[t \cdot \frac{j}{2k} e^{-j2kt} \right]_0^{\pi}}_{\text{boundary term}} - \underbrace{\int_0^{\pi} \frac{j}{2k} e^{-j2kt} dt}_{\text{remaining integral}}$$

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$$\pi \cdot \frac{j}{2k} \underbrace{e^{-j2k\pi}}_{=1} - 0 = \frac{j\pi}{2k}$$

Remaining integral:

$$\frac{j}{2k} \left[\frac{e^{-j2kt}}{-j2k} \right]_0^{\pi} = \frac{j}{2k} \cdot \frac{\overbrace{e^{-j2k\pi}}^{=1} - 1}{-j2k} = \frac{j}{2k} \cdot \frac{0}{-j2k} = 0$$

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So $\int_0^{\pi} t e^{-j2kt} dt = \frac{j\pi}{2k} - 0 = \frac{j\pi}{2k}$, and multiplying by $\frac{1}{\pi}$:

$$c_k = \frac{j}{2k} \quad (k \neq 0)$$

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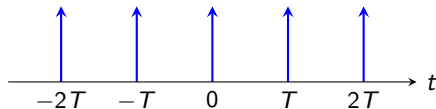
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Check (conjugate symmetry): since $x(t)$ is real, we need $c_{-k} = c_k^*$.

$$c_{-k} = \frac{j}{-2k} = -\frac{j}{2k}, \quad c_k^* = \left(\frac{j}{2k}\right)^* = -\frac{j}{2k} \quad \checkmark$$

1.5 – Impulse train

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The analysis integral can be taken over *any* single period, so pick $(-\frac{T}{2}, \frac{T}{2})$, which contains exactly one impulse (at $t = 0$):

$$c_k = \frac{1}{T} \int_{-T/2}^{T/2} \delta(t) e^{-jk\omega t} dt = \frac{1}{T} e^0$$

$$c_k = \frac{1}{T} \text{ for every } k$$

A flat spectrum: an impulse train contains every harmonic with exactly equal weight.

1.6 – Oppenheim 3.8: setting up

Given: $x(t)$ real and odd, period $T = 2$ ($\omega = \pi$), Fourier coefficients $a_k = 0$ for $|k| > 1$, and average power $\frac{1}{2} \int_0^2 |x(t)|^2 dt = 1$.

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- Real + odd (by 1.2) $\Rightarrow a_k$ purely imaginary and $a_{-k} = -a_k$. Also a_0 must be real *and* imaginary $\Rightarrow a_0 = 0$ (consistent with any odd signal having zero average).
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Write $a_1 = j\beta$ (β real, since a_1 is purely imaginary). Then

$$x(t) = j\beta e^{j\pi t} - j\beta e^{-j\pi t} = j\beta (e^{j\pi t} - e^{-j\pi t}) = j\beta \cdot 2j \sin(\pi t) = -2\beta \sin(\pi t)$$

So $x(t) = A \sin(\pi t)$ for a single real amplitude $A = -2\beta$ – the only freedom left is the value of A .

1.6 – Proof of Parseval's relation

Write $x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega t}$ and its complex conjugate $x(t)^* = \sum_{m=-\infty}^{\infty} a_m^* e^{-jm\omega t}$. The average power over one period is

$$\frac{1}{T} \int_T x(t) x(t)^* dt = \frac{1}{T} \int_T x(t) \left(\sum_m a_m^* e^{-jm\omega t} \right) dt$$

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Swap the sum and the integral:

$$\begin{aligned} &= \frac{1}{T} \sum_m a_m^* \underbrace{\int_T x(t) e^{-jm\omega t} dt}_{= Ta_m, \text{ by Eq. (3)}} \\ &= Ta_m, \text{ by Eq. (3)} \end{aligned}$$

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Since $a_m^* a_m = |a_m|^2$:

$$\boxed{\frac{1}{T} \int_T |x(t)|^2 dt = \sum_{k=-\infty}^{\infty} |a_k|^2}$$

1.6 – Using Parseval's relation

Parseval: average power = $\sum_k |a_k|^2$. Here only $k = \pm 1$ contribute, and $|a_{-1}| = |a_1| = |\beta|$:

$$1 = \frac{1}{2} \int_0^2 |x(t)|^2 dt = |a_1|^2 + |a_{-1}|^2 = 2\beta^2 \implies \beta = \pm \frac{1}{\sqrt{2}}$$

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Direct check: for $x(t) = A \sin(\pi t)$, $\frac{1}{2} \int_0^2 A^2 \sin^2(\pi t) dt = \frac{1}{2} A^2 \cdot 1 = \frac{A^2}{2}$ (since $\sin^2(\pi t)$ averages to $\frac{1}{2}$ over its period $T = 2$). Setting $\frac{A^2}{2} = 1$ gives $A = \pm\sqrt{2}$ ✓. These two signals are the only ones satisfying every condition – Parseval fixes the *magnitude* of A , but not its sign (phase), hence two valid answers.

2.1 – Product of signals $\rightarrow$ convolution of coefficients

Let $x(t) = \sum_k c_k e^{jk\omega t}$ and $y(t) = \sum_l d_l e^{jl\omega t}$ (same period T , $\omega = 2\pi/T$). Multiply:

$$x(t)y(t) = \sum_k \sum_l c_k d_l e^{j(k+l)\omega t}$$

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The bracketed quantity is exactly the m -th Fourier coefficient of $z(t) = x(t)y(t)$:

$$e_m = \sum_{k=-\infty}^{\infty} c_k d_{m-k} = (c * d)[m]$$

a **discrete convolution** of the two coefficient sequences – the dual of the (unrelated) fact that convolution in time corresponds to multiplication of Fourier coefficients... here it's the other way around: multiplication in time, convolution in frequency.

2.2 – The square wave studied in class: deriving c_k

Recall the standard bipolar square wave used in lecture: period T , $\omega = 2\pi/T$,

$$x(t) = \begin{cases} 1, & 0 < t < T/2 \\ -1, & T/2 < t < T \end{cases}$$

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c_k ($k \neq 0$): split the integral at the jump,

$$c_k = \frac{1}{T} \left[\int_0^{T/2} e^{-jk\omega t} dt - \int_{T/2}^T e^{-jk\omega t} dt \right]$$

2.2 – The square wave studied in class: result

Evaluating each piece:

$$\int_0^{T/2} e^{-jk\omega t} dt = \frac{e^{-jk\pi} - 1}{-jk\omega} = \frac{1 - (-1)^k}{jk\omega}, \quad \int_{T/2}^T e^{-jk\omega t} dt = \frac{1 - (-1)^k}{-jk\omega}$$

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Since $1 - (-1)^k$ is 2 for odd k and 0 for even k , and $1/j = -j$:

$$c_k = \begin{cases} \frac{-2j}{k\pi}, & k \text{ odd} \\ 0, & k \text{ even} \end{cases} \quad (k \neq 0), \quad c_0 = 0$$

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Setting $T = 2\pi$ ($\omega_0 = 1$) gives the version with period 2π :



2.2(a) – Centring the square wave (time shift)

We want $x_a(t) = x(t + \frac{\pi}{2})$ – shifting the class square wave left by $\frac{\pi}{2}$ makes it symmetric (even) about $t = 0$: $x_a(t) = 1$ for $|t| < \frac{\pi}{2}$, -1 for $\frac{\pi}{2} < |t| < \pi$.

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For odd k , $e^{jk\pi/2} = j^k$, which cycles $j, -j, j, -j, \dots$ as $k = 1, 3, 5, 7, \dots$. Working out $-2j \cdot j^k / (k\pi)$ term by term gives a real, alternating-sign result:

$$d_k = \frac{2(-1)^{(k-1)/2}}{k\pi} \quad (k \text{ odd}), \quad d_k = 0 \quad (k \text{ even}), \quad d_0 = 0$$

As expected from 1.2: $x_a(t)$ is real and even, so d_k came out *real* (the imaginary j 's cancelled) – a good consistency check on the shift.

2.2(b) – Shifted and sped up (time scaling)

$$x_b(t) = \begin{cases} 1, & -\frac{\pi}{4} < t < \frac{\pi}{4} \\ -1, & \frac{\pi}{4} < t < \frac{3\pi}{4} \end{cases} \quad (\text{period } \pi)$$

2.2(b) – Shifted and sped up (time scaling)

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Compare with $x_a(2t)$: high for $|2t| < \frac{\pi}{2} \Leftrightarrow |t| < \frac{\pi}{4}$, low for $\frac{\pi}{2} < |2t| < \pi \Leftrightarrow \frac{\pi}{4} < |t| < \frac{\pi}{2}$... matching the period- π pattern exactly. So $x_b(t) = x_a(2t)$ – a pure time-scaling by $\alpha = 2$, no further shift needed.

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$$e_k = d_k = \frac{2(-1)^{(k-1)/2}}{k\pi} \quad (k \text{ odd}), \quad e_k = 0 \quad (k \text{ even})$$

This is the whole point of the properties: no new integral was needed for either (a) or (b) – just a shift and a rescaling of the original class example.

2.3 – Oppenheim 3.5: setting up

Given $x_1(t)$ periodic with fundamental frequency ω_1 , coefficients c_k , and

$$x_2(t) = x_1(1 - t) + x_1(t - 1).$$

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$$\boxed{\omega_2 = \omega_1}$$

Coefficients: expand directly. Since $x_1(t) = \sum_k c_k e^{jk\omega_1 t}$,

$$x_1(1 - t) = \sum_k c_k e^{jk\omega_1(1-t)} = \sum_k [c_k e^{jk\omega_1}] e^{-jk\omega_1 t} \stackrel{k \rightarrow -k}{=} \sum_k [c_{-k} e^{-jk\omega_1}] e^{jk\omega_1 t}$$

so $x_1(1 - t)$ contributes coefficient $c_{-k} e^{-jk\omega_1}$ at harmonic k .

2.3 – Result

Similarly, $x_1(t-1) = \sum_k c_k e^{jk\omega_1(t-1)} = \sum_k [c_k e^{-jk\omega_1}] e^{jk\omega_1 t}$, contributing coefficient $c_k e^{-jk\omega_1}$ at harmonic k .

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$$b_k = c_{-k} e^{-jk\omega_1} + c_k e^{-jk\omega_1} = (c_k + c_{-k}) e^{-jk\omega_1}$$